Le Nguyen Gia Hung

AI undergraduate researcher | Explainable AI & multimodal learning

Research interests: time-series forecasting, transparency in AI and computer vision.

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Portfolio

Education

FPT University — B.Eng. in Artificial Intelligence 2023 – Dec 2027 (expected)
GPA: **8.62/10.0**. Certificate of Merit: Honorable Student of Trimester, semesters 3, 4, 5 and 6.

Publications & Manuscripts

Peer-reviewed conference publication

Gia-Hung Nguyen Le, Gia-Bao Pham Hoang, Tan-Phat Vo, Thu Le and Nhu Nguyen. "Proactive Air Quality Forecasting and Health Alert System for Melbourne." *AJCAI 2025, Springer LNAI*. First and corresponding author. DOI: 10.1007/978-981-95-4969-6_34.

Manuscripts under review

"Stress-Testing Multi-Source Cyanobacterial Bloom Forecasting under Sparse Monitoring." *Under review, AJCAI 2026*. First and corresponding author.

"Warning Before the Evening Peak: Health-Centred Air Quality Alerts in New Zealand and Australia." *Under review, AJCAI 2026*. Second author.

"Trustworthy Multimodal Anomaly Detection in Time Series with Conformal Calibration and Counterfactual Attribution." *Under review, ACM SIGKDD 2027*. Second author.

Selected Research Experience

SmokeNet: Air Quality Forecasting Jun – Nov 2025
First & Corresponding Author; published at AJCAI 2025 Academic research

- Designed a Transformer with seasonal embeddings and reconstruction loss to forecast Melbourne's $PM_{2.5}$; built chronological evaluation on 4+ years of OpenWeather data.
- Reported MAE **0.7470**, R^2 **0.9545** and **35.9% lower MAE than XGBoost** on the 2024 held-out evaluation.
- Evaluated the May 2024 smoke event separately: **57.7% lower MAE than the baseline** in that backtest.
- Inspected forecasts with SHAP, Integrated Gradients and DiCE.

Cyanobacterial Bloom Forecasting Jan 2026 – Present
First & Corresponding Author; AJCAI 2026 under review Academic research

- Built a causal weekly benchmark across **7 New Zealand lakes**: $n = 5{,}432$, with an untouched 2023–2024 holdout ($n = 170$).
- 7-day Chl-a history XGBoost reached weighted F1 **0.791** (+0.103 over full fusion, $p = 0.016$); 14-day full fusion reached **0.703**.
- Led manuscript writing and the 8-gate provenance audit: 22 apparent episode starts reduced to 3 eligible incidents.

Health-Centred Air Quality Alerts (NZ & Australia) Jan 2026 – Present
Second Author; AJCAI 2026 under review Academic research

- Evaluated exposure coverage across Canterbury's 10-year ($n = 620{,}057$) and Melbourne's 4-year ($n = 823{,}009$) hourly records.
- Reported that daily-mean triggers overstate exposure protection by 7.7–18.6 points; 42% of exposure occurred on sub-threshold daily-mean days.
- Built an archived weather-forecast panel model: 1-day-ahead MAE **3.19** $\mu g/m^3$, R^2 **0.54**.

Trustworthy Multimodal Anomaly Detection Jan 2026 – Present
Second Author; ACM SIGKDD 2027 under review Academic research

- Audited look-ahead text leakage across **9 domains**: a reported 4.52% error reduction from naive text fusion reversed under embargo (leakage fraction 1.50).
- Developed C-DLE with split-conformal calibration and Mondrian pools for false-alarm control under missing modalities.
- Designed cross-modal counterfactual attribution; reported **94.3% accuracy** against ground-truth anomalies with a **10% coreset**.

MERR-GAT: Multimodal Emotion Recognition Oct 2025 – Present
Second & Corresponding Author; in progress Academic research

- Develop emotion recognition from speech and text, using **WavLM + Compact Conformer**, **BERT/RoBERTa** and dual-channel graph-attention fusion.
- Inspect modality contributions with attention plots, SHAP and Integrated Gradients; attention alone is not a causal explanation.
- Evaluation is in progress on **IEMOCAP, RAVDESS and MELD**, using WA, UA and Macro-F1.

Vietnamese Speech Recognition Sep 2025 – Present
Co-Researcher & Co-Author; in progress Academic research

- Develop a **5.4M-parameter** Vietnamese ASR model with a 4-layer CNN, 3-layer residual BiLSTM and CTC training on **745 hours** of speech.
- Reported **WER 33.09%** and **CER 15.28%**; compare against PhoWhisper to study the trade-off between accuracy and model size. Evaluation notes.

PPE Detection (YOLOv8) Course project
Developer Computer vision

- Built 9-class detection, safe/unsafe compliance logic and a Streamlit webcam interface with violation reporting.

Leadership & Cloud Learning

SpeedyLabX Student Research Group Jan 2025 – Present
Founder & Student Lead FPT University

- Lead a 10-member undergraduate group, with weekly technical workshops, paper reading and shared conference-submission workflows; research spans 3 pilots and 4 manuscripts.

First Cloud AI Journey Program 2026
FCAJ Cloud Intern, Team NeuraX AWS Vietnam program

- Documented cloud learning, workshops and project work in the program journal.

Skills

Programming: Python, SQL/PostgreSQL; TypeScript, React, FastAPI.
ML & evaluation: PyTorch, Transformers, TensorFlow/Keras, Scikit-learn, OpenCV; SHAP, Integrated Gradients, DiCE, chronological cross-validation.
Data & tools: Pandas, NumPy, Matplotlib, Weights & Biases, Jupyter, Git, Docker, Linux, Streamlit.
Technical writing: manuscripts, documentation and blog posts.

Certifications

DeepLearning.AI: Natural Language Processing.
Coursera: Gradient to Production: MLOps & Model Serving.
IBM: Full Stack Software Developer; Data Science Fundamentals with Python and SQL; Statistics for Data Science with Python; Introduction to Containers with Docker, Kubernetes & OpenShift.
Packt: AI Engineer Professional.